

Batch and Streaming ETL

1. What is ETL?

ETL stands for:

None

Extract → Collect data from source systems

Transform → Clean / process / enrich data

Load → Store into destination system

Used to move data from operational systems into analytics/reporting systems.

2. Basic Flow

None

Database / APIs / Logs / Files



Extract



Transform



Data Warehouse / DB / Data Lake

3. What is Batch ETL?

Batch ETL processes data in groups at scheduled times.

Examples:

None

Every hour

Every night

Every month

Manual trigger

Data is collected first, then processed later.

4. Batch Example

Generate monthly bills:

None

All customer transactions of April

↓

Run job on May 1

↓

Generate PDF invoices

5. What is Streaming ETL?

Streaming ETL processes events continuously as soon as data arrives.

None

Event occurs



Immediately processed



Stored / forwarded

Useful for real-time systems.

6. Streaming Example

None

User clicks ad



Kafka receives event



Flink updates click counter instantly

7. Batch vs Streaming

Feature	Batch ETL	Streaming ETL
Processing	Periodic	Continuous
Latency	High	Low
Cost	Lower sometimes	Higher
Complexity	Simpler	More complex
Use Case	Reports, billing	Real-time analytics

8. Polling vs Interrupt Analogy

None

Batch = Polling

Streaming = Interrupt

Batch runs on schedule whether new data exists or not.

Streaming runs when new event arrives.

9. Real Example

Batch:

None

Run every midnight:

Import sales data

Update dashboard

Streaming:

None

Every sale instantly updates dashboard

10. Common Batch Tools

None

Apache Airflow

Luigi

Cron Jobs

AWS Glue

Used for scheduled workflows.

11. Common Streaming Tools

None

Apache Kafka

Apache Flink

Apache Spark Streaming

Apache Flume

Scribe

Used for continuous pipelines.

12. What is DAG in ETL?

ETL pipelines are often built as a **DAG**:

None

$A \rightarrow B \rightarrow D$

$A \rightarrow C \rightarrow D$

Where:

None

A = Extract raw files

B = Clean data

C = Validate data

D = Load warehouse

No circular dependency.

13. What is a Job?

A **Job** = one execution of pipeline.

Example:

None

Nightly ETL started at 1:00 AM

Completed at 1:22 AM

That run is one job.

14. Why Streaming Can Be Better

Instead of processing month-end huge data:

None

1 TB billing data on last day

Process daily/real-time:

None

Small chunks processed continuously

Benefits:

None

Lower month-end load

Faster reports

Easy debugging

Even resource usage

15. Challenges in Batch

None

Long-running jobs

Retry failures expensive

Late reports

Large compute spikes

16. Challenges in Streaming

None

Exactly-once handling

Duplicates

Ordering issues

Checkpointing

Higher complexity

Continuous monitoring

17. Best HLD Pattern

None

OLTP DB → Kafka → Stream Processor → Redis Dashboard

↓

Data Lake / Warehouse

Real-time + historical analytics together.

18. When to Use Batch

Use batch for:

None

Billing

Payroll

Nightly reports

Large migrations

CSV exports

Backups

19. When to Use Streaming

Use streaming for:

None

Fraud detection

Live dashboards

Monitoring

Recommendations

Alerting

IoT metrics

20. Interview Talking Points

If asked ETL design:

None

Use Batch for heavy scheduled processing

Use Streaming for low-latency systems

Use Kafka + Flink for real-time

Use Airflow for orchestration

Store final data in warehouse

21. One-Line Summary

Batch ETL processes data periodically in chunks, while Streaming ETL processes data continuously in real time. Modern scalable systems often use both together.